Railtrack Company Code of Practice

METHODOLOGY FOR THE DEMONSTRATION OF COMPATIBILITY WITH FS2600 TRACK CIRCUITS ON THE D.C. RAILWAY

ENDORSEMENT & AUTHORISATION

Endorsed by:

A Simmons, Professional Head of Signal Engineering

Accepted for issue by:

P McAleer, Acting Head of Business Standards

Authorised by:

A McNaughton, Chief Engineer

ISSUE RECORD

ISSUE	DATE	COMMENTS
1	**February 2003**	**New Code of Practice**

IMPLEMENTATION

The provisions of this code of practice may be used from the date of issue.

DISCLAIMER

SUPPLY

Copies of documents are available electronically, within Railtrack's organisation. Hard copies of this document will be available to Railtrack staff on request to the Document Controller and to other organisations from Technical Indexes Ltd (01344 404409).

Published & Issued by **Railtrack PLC (part of the Network Rail group of companies)**
Railtrack House
Euston Square
LONDON
NW1 2EE

Contents

1.0 Objectives

1.1 Purpose

The purpose of this document is to provide a methodology to demonstrate compatibility with FS2600 track circuits on Railtrack 750V d.c. electrified railway.

1.2 Scope

This document covers FS2600 track circuits installed in double rail configuration on d.c. electrified Railtrack controlled infrastructure.

Methodology is based on level alone for simplicity of demonstration. This is achieved by:

- Providing a characterisation of the track circuit components.
- Providing a characterisation of the infrastructure.
- Providing a methodology to define the susceptibility for worst case scenarios.

Combining these elements will allow an analytical investigation to find the maximum permissible interference for trains in d.c. electrified areas, with double rail FS2600 circuits applicable to Railtrack infrastructure. The minimum levels found from the susceptibility analysis are taken as the maximum permissible interference levels. Methods are included to analyse intact and faulty infrastructure for single and double track configurations.

The FS2600 track circuit has been designed to be 'traction immune', so that it cannot be energised by any single interference frequency or transient. This is achieved by using Digital Signal Processing (DSP) to monitor the signal it receives from the track. The use of DSP enables the validity of the signal to be checked in terms of its frequency content and amplitude at the same time. This overcomes the traditional problem encountered with separate level and code checking systems which can be erroneously activated by small genuine signals with a higher general noise signal overlaid. Energisation of the receiver requires a train to produce a modulated current which gives an FFT signature similar to that of the FS2600 transmitter. This is unlikely to occur in practice. Hence the results of this analysis are highly pessimistic.

1.3 Scenarios covered by this Document

A FS2600 track circuit will continue to operate in the presence of interference currents flowing in the traction return system of any single track circuit up to the levels indicated in Table 1 [12]. The system will also withstand up to 60Vrms across the rails at the receiver end of the track circuit without failure.

Frequency (Hz)	Maximum Current in Traction Return Rails (A)
DC	8,000
1 to 152	400
152 to 312	100
312 to 576	3
576 to 3,000	100

Table 1: FS2600 Traction Interference Reliability Levels

The tolerated interference current level for the FS2600 frequency band for RSF are higher than those allowed for wrong side failure (WSF) [12], hence this documents stresses on failure modes causing WSF even if the probability of these happening is considered incredible.

- Failure Mode 1 — Wrong side failure caused by interference current with all rails intact
- Failure Mode 2 — Wrong side failure caused by interference currents, with a broken traction return rail or impedance bond connection

- Axle to Axle Voltage — Wrong side failure caused by a train on an adjacent track producing axle to axle voltage

The analysis was carried out for single and two track areas only. The results produced are valid for areas with any number of tracks, as the effect of interference is reduced by the increasing number of parallel return paths.

Only two interference mechanisms (return current through track circuit and axle-to-axle voltage) are analysed. Other mechanisms, such as induction directly into the impedance bond or receiver tail cables are not covered by this report, since very high interference-field levels would be required to have a significant effect.

2.0 SOURCE DOCUMENTS

2.1 Group and Line Standards

1. Electrified Lines Traction Bonding, GL/RT1254, Issue 1, April 2000.
2. Railtrack Company Product Specification - Westinghouse Signals FS2600 Track Circuits, RT/E/PS/11760, Issue 1, December 2000.
3. Railtrack Company Product Specification - Compatibility between Electric Trains and Electrification Systems, RT/E/C/27010, Issue 1, June 1999.
4. Railtrack Company Product Specification - Design and Installation of Negative Bonding and Associated Equipment on D.C. Electrified Lines, RT/E/C/27022, issue 1, March 1998.
5. Railtrack Company Specification MRS - Index of Railtrack Documents Relating to Electrical Distribution & Electro-Mechanical Plant Engineering Activities, RT/E/S/20000, Issue 9, April 2001.

2.2 Previous Safety Case documentation

6. 'Class 92 Locomotive EMC Safety Assessment - Volume 6 - FS2600 Track Circuits', AEAT Rail (former TCI) Ref. 120-dr0344, Version b0, September 1997.
7. Class 365/5 Networker Express EE&CS EMC Safety Assessment for D.C. Operation, AEAT Rail (former TCI) Ref. TCS120-dr0148, September 1997.
8. 'Juniper Project - FS2600 Track Circuit Interference Limits - DC Routes', AEAT Rail Ref. PRJ-SCSE(JUN)-RP-6, Issue 2, October 1998.

2.3 Other Standards

9. prEN 50238 'Railway Applications: Compatibility between Rolling Stock and Train Detection Systems' Final draft 1998
10. Specification for the Design and Installation of Negative Bonding and Associated Equipment on the High Current D.C. Electrified Lines of Network South East, EHQ/SP/C/501, Issue 1, November 1994.

2.4 Other Documentation

11. FS2600 System Design Specification, Westinghouse Signals Limited X1306/35/A00006, issue 2, I J Lewis, 15 March 1994.
12. FS2600 Traction Compatibility Explanatory Information, Westinghouse Signals Ref. X1306/35/A00056.
13. FS2600 Track Circuit, Transfer Function for Interference Currents, British Rail Research STS-D-CTP(FST)-RP-2, issue 1, N B Rambukwella, 31 January 1994.
14. FS2600 Track Circuit Handbook, Westinghouse Signals Limited X1306/35/TCH M, issue 2, R Wood, 31 March 1994, AEA Technology Rail Ref: STS-D-CTP(FST)-ID-21, issue 4.
15. British Railway Track, Design, Construction and Maintenance, Permanent Way Institute, 5th Edition, 1979.

3.0 INFRASTRUCTURE ASSUMPTIONS

The infrastructure assumed during the analysis is as described below, further assumptions are listed in Appendix B:

1 The track circuits and cross-bonding is assumed to be set up in accordance with the FS2600 Track Circuit Handbook [14].

2 The interference source is a train capable of producing an FSK signal identical to that of the track circuit signal.

3 Ballast resistance is assumed to be infinite.

4 A balanced train shunt of 10 $\mu\Omega$ per wheel was used for double rail track circuits.

5 100% imbalance is assumed, for faulty infrastructure cases, this is pessimistic compared to the nominal imbalance values of 70%/30% in the case of 750 V d.c. power supplies [8].

6 The traction bonding arrangements are compliant with [1] and [4].

7 The running rails are assumed to be type BS113A, normal grade. Check rails are excluded. The sleepers are conventional concrete or wooden type. The track is assumed to be of standard gauge.

8 Impedance bonds' saturation effect is not considered.

9 The permissible interference current level is lower for operation at Channel-10 carrier frequency of 508.8 Hz [13]. The results presented reflect Channel-10 operation but are being applied across FS2600 band.

10 Impedance bonds are type 3 - WH3 (Westinghouse).

11 The infrastructure fault is always positioned to achieve the maximum effect on the track circuit receiver.

4.0 Abbreviations used

I_{TB}	Transmitter Breakthrough Current
I_{int}	Traction Interference Current
$I_{\mathrm{int\_max}}$	Maximum Permissible Interference Current
I_{RX}	Receiver Current
$I_{RX\,\mathrm{max}}$	Maximum Permissible Receiver Current, Track Circuit Occupied
I_{da}	Receiver Drop-away Current
V_{RX}	Receiver Voltage
V_T	Transmitter Voltage
V_{aa}	Axle-to-axle Voltage
I_{aa}	Current at the Receiver Caused by Axle-to-axle Voltage

V_{aa_max} Maximum Permissible Axle-to-axle Voltage

I_{aa_max} Maximum Permissible Receiver Current, Track Circuit Occupied and Axle-to-axle Voltage

5.0 FS2600 Track circuit operation

The system architecture of the double rail track circuits is shown in Figure 1. The FS2600 has only been approved in double rail configuration.

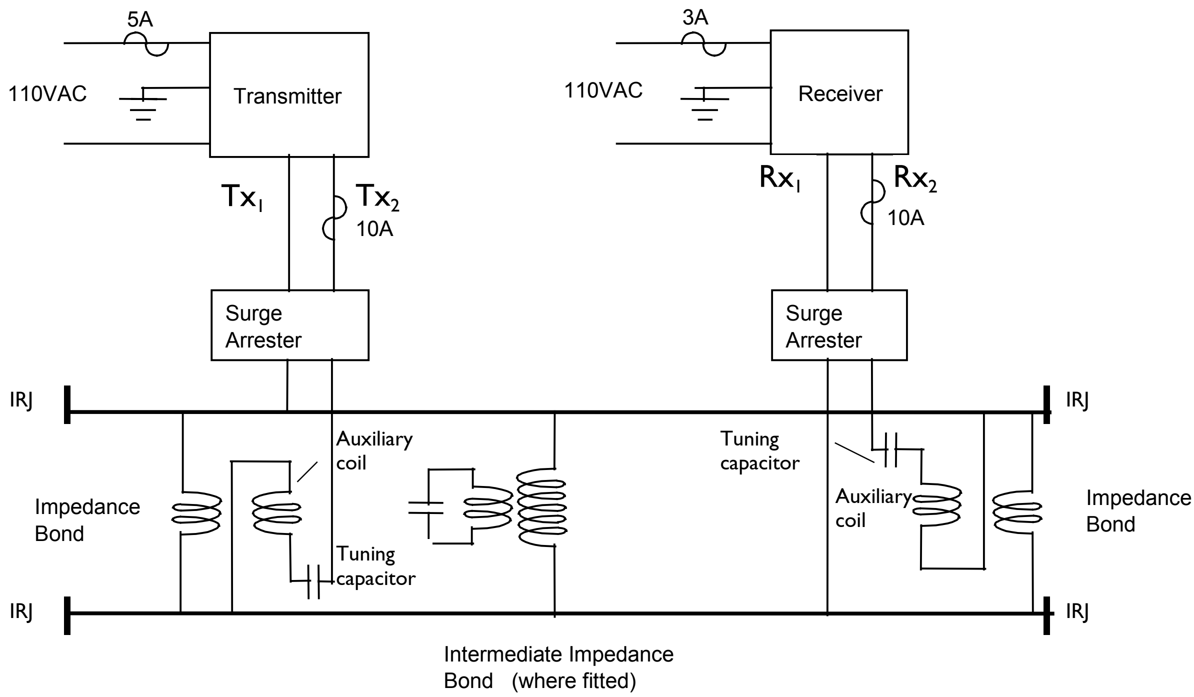


Figure 1: FS2600 track circuit in double rail configuration

The traction current by-passes Insulated Rail Joints (IRJs) via impedance bonds while the track circuit current is contained within the limits of the track circuit the extend of which is defined by the IRJs. The track circuit signal from the transmitter is passed to the rails via an auxiliary coil on the impedance bond. A fixed value tuning capacitor in the impedance bond is used to tune the auxiliary coil of the impedance bond to the operating frequency of the track circuit, i.e. the carrier frequency.

The signal in the rails flows via the impedance bond at the receiver end. This current results in a voltage being induced in the auxiliary winding which is connected to the receiver. A tuning capacitor is used to tune the bond to the operating frequency of the track circuit. In the double rail configuration FS2600 track circuits may be installed between 50m and 1200 m in length.

Protection against EMI is provided by the FSK nature of the transmitted signal which must be present for a minimum of 944 milliseconds for the receiver to recognise it as a valid transmitted signal.

The transmitter generates a FSK signal which is fed to the rails at one end of the track circuit. The transmitter output is derived from three crystal oscillators. One provides a low frequency modulation signal, the other two providing an upper and lower shift frequency. The modulation frequency is used to switch the output between the upper and lower shift frequencies. Thus the FSK waveform consists of the upper shift frequency (upper side band) alternating with the lower shift frequency (lower side band) at a rate determined by the modulation frequency. If such an FSK signal is observed in the frequency domain for periods of time much less than the period of the modulation frequency then alternate "bursts" of upper and lower shift frequency will be seen (F_{upper} and F_{lower}). Ten pairs of frequencies are used in the range of 371.8 - 525.8 Hz, as shown in Table 2 below:

Channel	F_{lower}	F_{upper}	Lower Side Band (Hz)	Carrier (Hz)	Upper Side Band (Hz)
1	371.8	405.8	373.2	388.8	404.4
2	386.2	420.2	387.6	403.2	418.8
3	400.6	434.6	402.0	417.6	433.2
4	415.0	449.0	416.4	432.0	447.6
5	424.6	458.6	426.0	441.6	457.2
6	439.0	473.0	440.4	456.0	471.6
7	453.4	487.4	454.8	470.4	486.0
8	467.8	501.8	469.2	484.8	500.4
9	477.4	511.4	478.8	494.4	510.0
10	491.8	525.8	493.2	508.8	524.4

Table 2: FS2600 Track Circuit Signal Frequencies

The important features in respect of the track circuits immunity to interference are as follows:

1. The FS2600 track circuits operate by the transmission of two separate frequencies termed F_{lower} and F_{upper}, which are modulated (i.e. switching between the two frequencies F_{lower} and F_{upper}) at a rate of 15.6 Hz.
2. The receiver contains a Digital Signal Processor (DSP) to analyse the received signal. The nature of this signal processing is such that the carrier frequency of the FSK signal must be an integer multiple of 4.8 Hz. The receiver will reject signals as they approach +/- 2.4 Hz from their nominal value. Since all signals are crystal controlled they will be within 400 parts per million of their nominal value.
3. The receiver performs sampling and analysis of the input signal in parallel. The sampling process lasts for 208.3 ms and the analysis takes 236 ms. Figure 2 depicts the relationship between these two tasks:

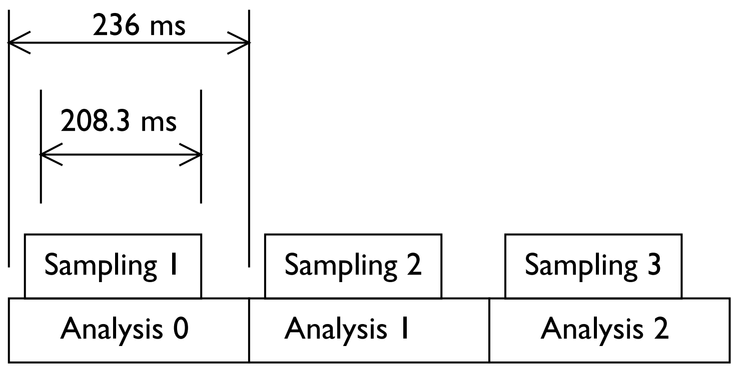


Figure 2: Cycle of Operation

The sampling frequency used is 614.4 Hz, so 128 samples of the input signal are obtained per sampling period. The input signal is heterodyned and filtered prior to being sampled. The sequence of operations performed by the receiver is detailed in [12].

The analysis conducted by the receiver on the transmitted FSK signal consists of a total of six tests for each 208.3 ms sample of the input signal. The six tests carried out are as follows:

- a check to ensure that the power level of the carrier frequency is greater than that of both side bands,
- a check that the power level of the carrier frequency is above the 'upper hysteresis' level in which case it is passed. If the power level of the carrier frequency is below the 'lower hysteresis' level the test is failed. If the power level of the carrier frequency is between the upper and lower hysteresis levels then the track circuit will not change state,
- a check that the power level of the upper sideband remains between the upper and lower limits (corresponding to the 'on' and 'off' conditions),
- a check that the power level of the lower sideband remains between the upper and lower limits (corresponding to the 'on' and 'off' conditions),
- a check of the power levels of the two sidebands to ensure that the largest is less than four times as great as that of the smallest,
- a check that the relative phase of the resultant of the two main sidebands with respect to the carrier is 90° +/- 45°.

If all of the above six tests pass on four consecutive cycles of the receiver analysis (i.e. the signal is recognised as a valid transmitted signal) then the receiver output is energised. If any of these tests fail on any two consecutive cycles then the receiver output is de-energised.

In order for the receiver to recognise a signal as 'valid' a total of 24 tests must be carried out and passed (six tests on four consecutive sampled signals). Since it takes 236 ms for one set of tests (incl. sampling and analysis) this means that the signal must be present for a minimum of 944 ms for the track circuit to energise.

There is further time delay, implemented by the output relay within the receiver. The transmitter FSK signal must exist for a minimum of 1.29 seconds before the receiver will show the track clear (receiver pick-up time), see [7].

For double rail configuration impedance bonds are used. The methodology provided accommodates for type 3 - WH3 (Westinghouse). The use of type 3 impedance bonds rated at 4,500 A at all substations and track paralleling huts is mandated by [5] and [10]. The use of Howells type 3 impedance bonds has no impact on the analysis as the different parameters are compensated by different value tuning capacitors [2] to achieve the same effect.

6.0 Susceptibility Analysis

Sections 6.1, 6.2 and 6.3 show how the various components, which were used in the analysis, were modelled. Section 6.4 shows how these components were combined to produce the models used for analysis.

6.1 Track Circuit Components

The following circuit diagrams in
Figure ***3*** show how the track circuit transmitter, receiver and impedance bonds were modelled, using the values listed in APPENDIX A: Data Used for Modelling.

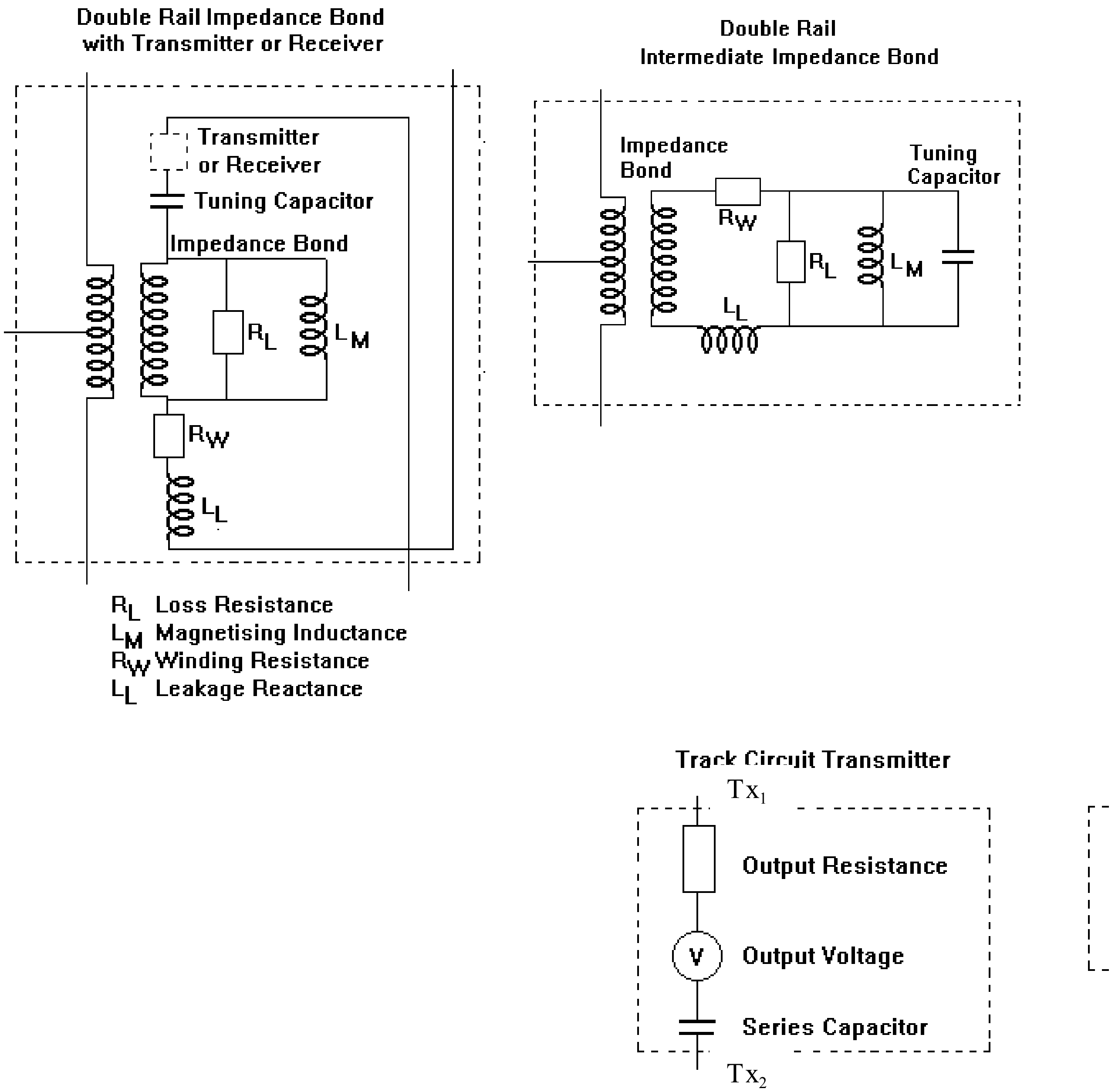


Figure 3: Model of track circuit elements

The surge arresters shown in Figure 1 are assumed to be open circuit and therefore are not included in the model.

6.2 Track

The track layout was modelled as shown in Figure 4. Only one third rail needs to be modelled, as in the analysis only one train was ever assumed to be on the track, drawing current, and therefore no current would flow in the other third rail. Currents produced by other trains in the area are not modelled, but are allowed for in the assessment margins.

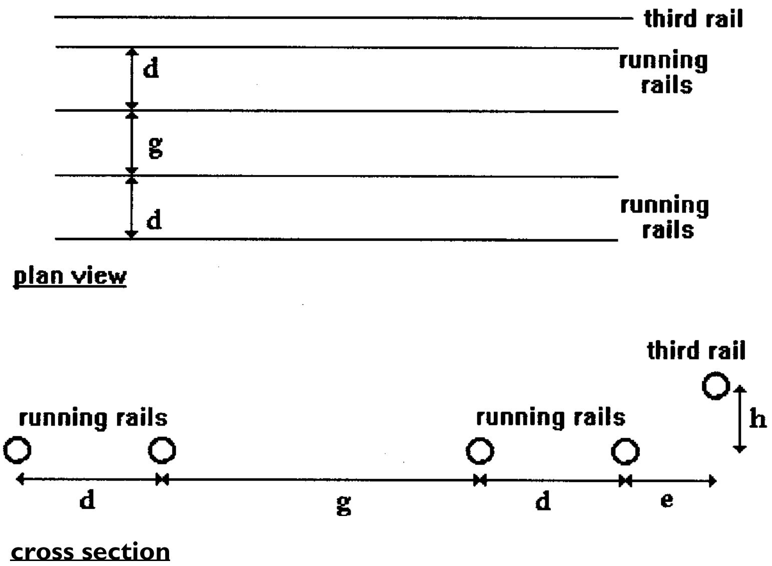


Figure 4: Track layout modelled

The dimensions are assumed as shown below:

d	1.435 m	See Reference 11, page 21.
g	1.83 m	Assumed to be approximately 6 feet.
h	0.076 m	See Reference 3, page 23.
e	0.37 m	Calculated from values found in and Reference 15, page 30 (rail dimension only).

Perimeter of running rail	0.63 m	Assumed to be BS113A, normal grade. See Reference 16.
Perimeter of conductor rail	0.58 m	See Reference 16.

Based on these dimensions, values of external self inductance and mutual inductance were calculated using AEAT Rail methodology, [17]. Values for measured rail resistance and internal self-inductance (inductance due to flux linkage within a rail itself) from [16] are used in the model. The values used for the parameters in the model are listed in APPENDIX A: Data Used for Modelling. Two models were created, of 10 and 600 m lengths, equal to the minimum allowable cross-bond spacing from [14] and the maximum allowable length double rail track circuit with a low output transmitter setting [2] correspondingly.

The schematics of the track circuit models in APPENDIX B: Track Circuit Schematic accommodate for the values of external and mutual inductance of the rails.

The effects of ballast resistance and inter-rail capacitance were ignored.

The cross bonds are modelled as 10 μΩ resistors, connected between the relevant rails or impedance bonds. When the cross-bonds are 'removed', to simulate single track, the value of the 10 μΩ resistors is increased to 1000 MΩ (effectively open circuit).

6.3 Train

6.3.1 Traction Return Current Model

The train occupying the track circuit under consideration is modelled as a point source. The shunt of this train is modelled using two series resistors between the rails, as shown in Figure 5:

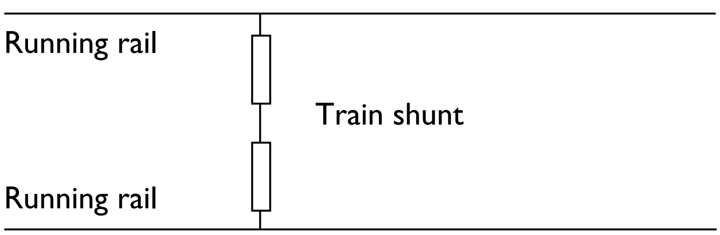


Figure 5: Model of train producing interference current

Since the track circuit is double rail traction current will return via both wheels / rails, a balanced train shunt is used. Both resistors are 10 μΩ.

In double rail track circuits the traction current will flow from the wheel set to the rails as shown in Figure 6.

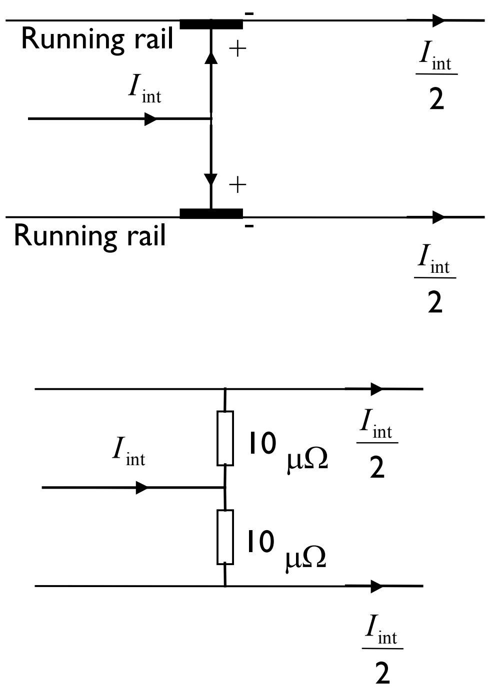


Figure 6: Double rail track circuit, current flow and assumed model of train shunt

The interference source (the train) is placed between the third rail and the middle of the train shunt. This is modelled as a current source producing current at 508.8 Hz (channel 10 track circuit frequency).

6.3.2 Axle-to-axle Voltage Model

The train producing the axle-to-axle voltage is modelled an ideal voltage source between the impedance bonds, producing a voltage at 508.8 Hz.

6.4 Complete Track Circuit Models

6.4.1 Most Sensitive Receiver Setting

The track circuits (short and long) are set up as per Reference 14, TCH M70, Section 3.2. To determine the effect of transmitter breakthrough the transmitter voltage was varied. The transmitter voltages used were 65 V for the short track circuit and 110 V for the long track circuit. This was done so that the receiver voltage across the rails could then be calculated using the same gain as was used for establishing the current produced by interference.

When finding the current produced at the receiver by interference current, the most sensitive gain was used, (50 mA flowing into the receiver corresponds to 3 V measured at the receiver across the rails, see Reference 11, Table 8). These give the nominal values. Breakthrough current is calculated from the minimum receiver pick-up current.

6.4.2 Short Double Rail

The model which was created for analysis of failure modes on short double rail track circuits is shown in Figure 7. In this diagram the train is shown close to the transmitter.

The length of track modelled was 600 m, this is the maximum allowable length double rail track circuit with a low output transmitter setting [2].

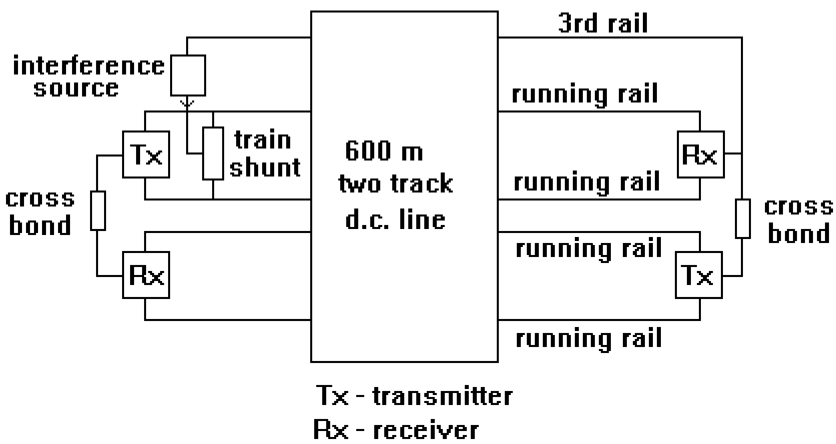


Figure 7: Short double rail track circuit, two track area, train near to transmitter

The adjacent track circuit is assumed to be Channel 1 as being the next most vulnerable to the one modelled [13] and because the adjacent track circuit is always of a different channel [2]. For the single track analysis the cross-bond resistors were increased to a high value.

6.4.3 Long Double Rail

The model used for analysis of failure modes on long double rail track circuits is shown in Figure 8. For simplicity, the intermediate impedance bond is placed right in the middle of the track circuit although the separation distance between impedance bonds can be from 10m to 800m [2]. In this diagram, the train is shown shunting the intermediate impedance bond.

The length of track modelled was 1200 m, as this is the maximum allowable length double rail track circuit with a high output transmitter setting [2].

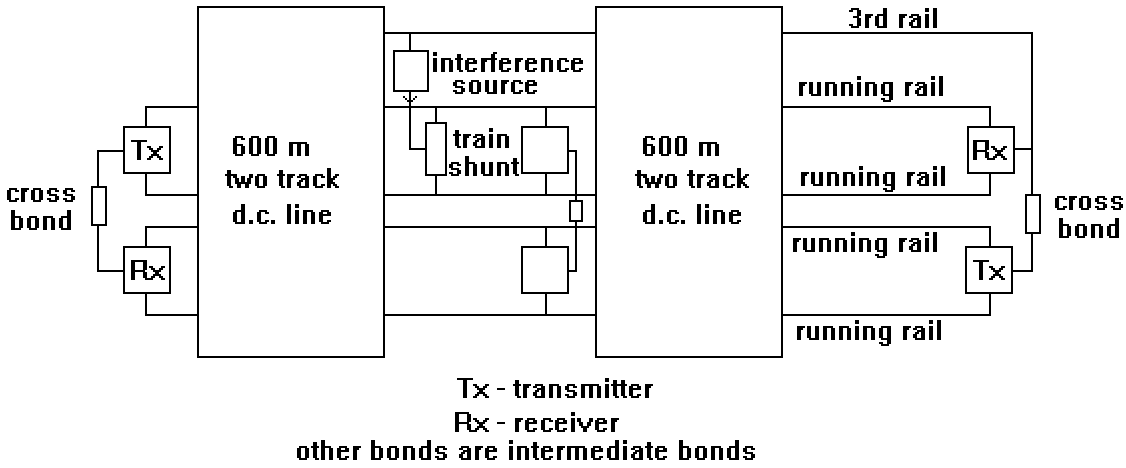


Figure 8: Long double rail track circuit, two track area, train shunting intermediate impedance bond

Tuning capacitors and magnetising inductances in the intermediate impedance bond are considered to be key parameters for sensitivity analysis, not the separation distance [8].

The adjacent track circuit is assumed to be Channel 1 for the same reasons given in 6.4.2. For the single track analysis the intermediate impedance bond was removed, as well as increasing the cross-bond resistors to a high value.

6.4.4 Axle-to-Axle Voltage Analysis

The model used for the analysis of axle-to-axle voltage is shown in Figure 9.

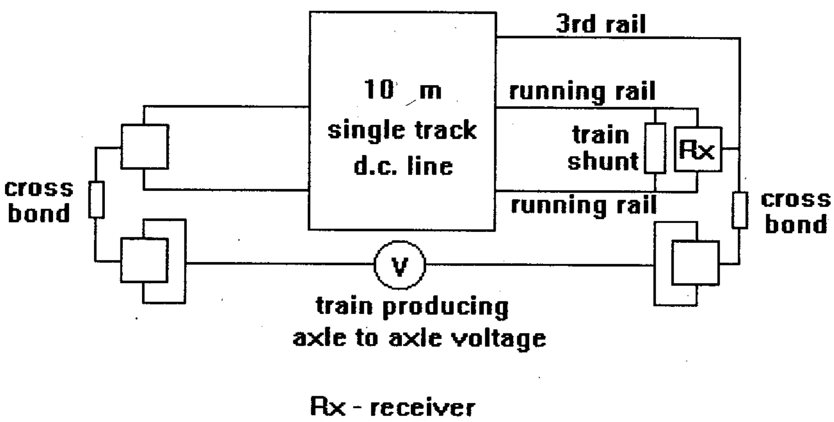


Figure 9: Double rail track circuit, model used for axle-to-axle voltage analysis

7.0 Modelling Methodology

7.1 Interference allowance

The maximum permitted interference was budgeted to allow for additional interference from other sources.

For intact infrastructure the allowance for the train interference on the track circuit is reduced to one third of the total interference budget and with failed infrastructure this is increased to one half.

The interference budget takes account of the following interference sources:

- Transmitter feed-through (from other track circuits of the same frequency – see below)
- Axle-to-axle voltage and conductive interference from trains on parallel tracks
- Trains remote from the track circuit but in the same feeder section
- General background noise from other sources

The level of transmitter feed-through is largely dependent on the train shunt impedance, and increases as the impedance rises. The methodology accommodates for this factor by using the most sensitive gain of the receiver (see 6.4.1). Any interference generated by the train must be at the same frequency, in phase, and modulated at the same rate as the transmitter for this to be a problem - this is most improbable in practice.

Feed-through from other track circuits is minimised by the design and layout of the track circuit [2]. The probability of the train producing interference that matches the feed-through in frequency, phase and modulation rate is assumed to be negligible.

The axle-to-axle voltage generated by a train on a parallel track can impact the normal operation of the track circuit via the cross bonds.

The contribution from trains remote from the track circuit will be small. These sources must be at the same frequency, in phase, and modulated at the same rate as interference from the train being analysed for this to be a problem - this is improbable.

General background noise from other sources may be considered insignificant, because of the requirement to match the frequency, phase, and modulation rate of the interference generated by the train.

7.2 Intact Infrastructure Case

Intact Infrastructure case considers the mechanism by which a wrong side failure can occur with intact infrastructure with train shunt across the rails.

The track circuits are double rail, therefore the traction current flows in both rails. The voltage across the receiver is caused by current imbalance in the running rails. The primary source of current imbalance is the mutual coupling between the running rails and the conductor rail. This is accommodated for in the detailed model in APPENDIX B: Track Circuit Schematics.

7.2.1 Wrong Side Failure

The interference at track circuit frequencies is assumed to come from the following sources:

- interference from the train being analysed;
- other trains in the area, including trains producing axle-to-axle voltage, where appropriate;
- background current from other sources.

Therefore a train interference allowance of a third is used. For this failure mode it is assumed an arithmetic addition of the interference currents. In practice it is unlikely that the interference sources would be producing the same signal, in phase.

This failure mode was analysed with track circuits at their maximum allowable length (i.e. 600 m for short double rail track circuits, 1200 m for long double rail track circuits). The basic models used in the analysis are shown in Figure 7 and Figure 8. The following scenarios were considered for single track and two track line in order to find the worst case.

For the track circuit analysis, the voltage at the receiver, V_{RX} was measured with transmitter voltage V_T set to the maximum values established in 0and then, V_T set to zero. The following calculations were made:

1. To quantify the transmitter breakthrough current, I_{TB} was calculated for transmitter voltage, V_T set to values established in 6.4.1 and train shunt resistance equal 20 $\mu\Omega$.
2. The interference generated current at the receiver, I_{RX} was calculated for transmitter voltage, V_T set to zero and train shunt resistance equal to 20 $\mu\Omega$.

For the track circuit to drop-away, the current in the receiver must fall below its drop-way level I_{da}. When the track circuit is occupied this current consists of two components, the one produced by the transmitter breakthrough, I_{TB} calculated from bullet point 1 above, and that due to interference from other sources, I_{int}. Only a proportion of the total rail current is coupled into the receiver (I_{RX}). This proportion is determined by track circuit model. For train shunt resistance equal 20 μΩ the transmitter feed through is practically zero thus the only current that flows through the receiver is that due to sources other than the track circuit transmitter, see Table 3 and Table 4. It is considered that the total currents at the receiver can be combined using root-sum-square addition. This is a valid summation rule since the interference sources are independent and the interference current will not be locked to the FSK of the track circuit transmitter, [8], [9]:

$$I_{da} = \sqrt{I_{RX\max}{}^2 + I_{TB}{}^2}$$

where $I_{RX\max}$ is the maximum permissible interference current.

$$\therefore I_{RX\max} = \sqrt{I_{da}{}^2 - I_{TB}{}^2}$$

The interference current generated at the receiver point is proportional to the traction interference current injected in the track circuit. Hence:

$$I_{RX} \propto I_{\text{int}} \text{ and } \mathrm{I}_{RX\max} \propto I_{\text{int\_max}}$$

where $I_{\text{int\_max}}$ is the maximum permissible traction interference current.

$$\Rightarrow \frac{I_{\text{int\_max}}}{I_{\text{int}}} = \frac{\sqrt{I_{da}{}^2 - I_{TB}{}^2}}{I_{RX}}$$

$$I_{\text{int\_max}} = \frac{I_{\text{int}}\sqrt{I_{da}{}^2 - I_{TB}{}^2}}{3I_{RX}}$$

The train is allowed to produce a third of the allowable interference current:

7.2.2 *Right Side Failure*

Interference currents generated by the train will flow back to the substation along the running rails. These currents can interfere with the operation of track circuits that are not occupied. If the waveform of the interfering current is identical to that of the track circuit current and is in phase opposition to it, then the resultant voltage at the receiver could be reduced below the drop-away value, causing a right side failure.

The magnitude of the interfering voltage at the receiver will depend, mainly on the current level and how the interference current is shared by the two running rails. In a 3rd rail system, the impedance to current flow is different for the two running rails causing a current imbalance. Since the impedance bond effectively acts as a three winding transformer it can support a current imbalance in the two halves of the primary, thus causing a receiver current to flow. The situation can be considered from the schematic for intact infrastructure in APPENDIX B: Track Circuit Schematics.

7.3 Faulty Infrastructure Case

In the case of double rail track circuits, the threshold to cause wrong side failure may be reduced by a break in either of the rails or in an impedance bond connection. Therefore these conditions need to be examined.

The breaks modelled for double rail track circuits are shown in Figure 10. Breaks at 10 positions were analysed individually for the long double rail two track case. Breaks 1 to 4 were analysed for both short and long double rail single track cases, since these do not have intermediate impedance bonds.

For each break, the train shunt was placed in the worst-case position either A, B or C in Figure 10.

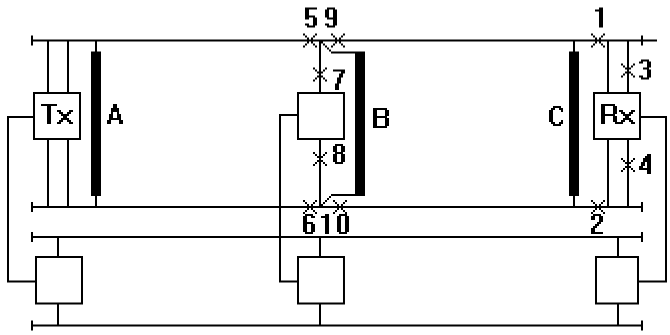


Figure 10: Positions of break and train modelled for double rail track circuits

Under infrastructure failure conditions it is assumed that the train under consideration can produce one half of the maximum allowable interference current, after taking the transmitter breakthrough into account.

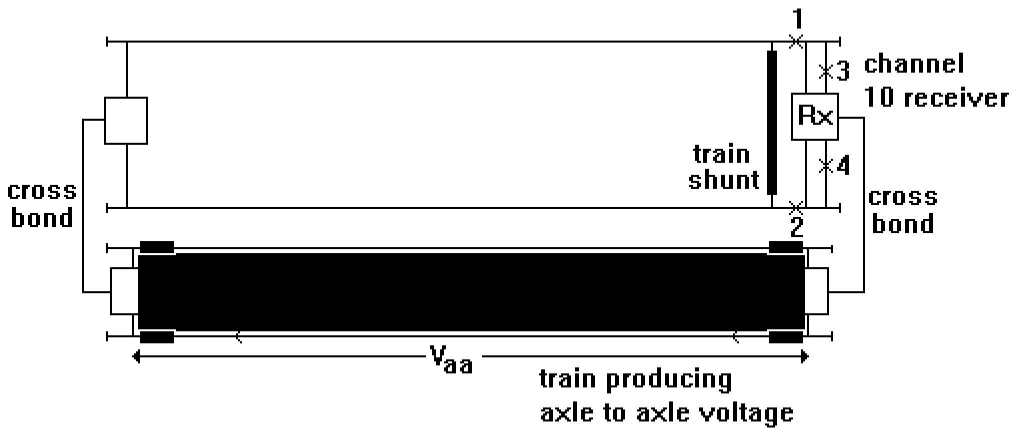
As in the Intact Infrastructure Case, section 7.2 the analysis was carried out for maximum length track circuits and for the same scenarios. The differences in analysis for Faulty Infrastructure case are:

- The 'train shunt' was placed at various positions;

- The infrastructure faults were introduced by using a high value resistor in the circuit at the appropriate position;

7.4 Axle to Axle Voltage

For axle-to-axle voltage to pose a risk to a track circuit an adjacent track there must be an infrastructure failure within that track circuit.



The situation considered for double rail track circuits is shown in Figure 11.

Figure 11: Wrong side failure caused by axle-to-axle voltage, double rail track circuit

The length of track modelled is 10 m, which is the minimum allowable cross-bond spacing [14]. The minimum cross-bond spacing is the absolute worst case . In this case the train shunt is near to the receiver and a break at one of four positions may cause the wrong side failure. Each of these breaks were modelled successively, as for Infrastructure Failure Case, section 7.3. The distance between the axles of the train producing the axle-to-axle voltage, is assumed to be the same as the distance between the cross-bonds. This train was modelled by a voltage source, as shown in Figure 9. The impedance-bonds are all intermediate bonds except the one connected to the receiver.

The transmitter breakthrough voltage was not modelled, but it was taken into account in the calculations. The value of transmitter breakthrough is assumed to be the same as that produced for Infrastructure Failure Case.

The current at the receiver was measured with the train producing an axle-to-axle voltage of V_{aa}. The results were then calculated as follows:

The maximum current allowable at the receiver, when there is a train occupying the track circuit is the drop-away current, I_{da} made up of the current produced by the train on the adjacent track, I_{aa} and the transmitter breakthrough current, I_{TB}. Hence

$$I_{da} = \sqrt{I_{aa\_max}{}^2 + I_{TB}{}^2}$$

where I_{aa_max} is the maximum permissible interference generated current.

$$\therefore I_{aa\_max} = \sqrt{I_{da}{}^2 - I_{TB}{}^2}$$

The interference generated current is proportional to the axle-to-axle voltage. Hence:

$$I_{aa} \propto V_{aa} \text{ and } \mathrm{I}_{\mathrm{aa\_max}} \propto V_{aa\_max}$$

where V_{aa_max} is the maximum permissible axle-to-axle voltage.

$$\Rightarrow \frac{V_{aa\_max}}{V_{aa}} = \frac{\sqrt{I_{da}{}^2 - I_{TB}{}^2}}{\mathrm{I}_{\mathrm{aa}}}$$

Maximum allowable interference to be produced by the train is a half of the total allowed:

$$V_{aa\_max} = \frac{V_{aa}\sqrt{I_{da}{}^2 - I_{TB}{}^2}}{2I_{aa}}$$

8.0 Worked Example And Results

Specified drop away voltage, V_{da} = 1.6 V (ref. 13)

Interference current injected in failure mode analysis, I_{int} = 5 A

Assumed drop away current for double rail track circuits , $I_{da} = V_{da} \times \frac{50}{3} = 26.7mA$

This is calculated using the methodology in 6.4.1.

8.1 Intact Infrastructure

Equation used for double rail, Intact Infrastructure analysis (from Section 7.2):

Maximum permissible interference current, $I_{int\_max} = \frac{5000\sqrt{26.7^2 - I_{TB}{}^2}}{3I_{RX}}$ mA

Number of Tracks	Receiver Current (mA)		I_{int_max} (mA)
	I_{TB}	I_{RX}	
1	0	51.9	857
2	0	54.9	810

Table 3: Short double rail track circuit, intact infrastructure results

Number of Tracks	Receiver Current (mA)		I_{int_max} (mA)
	I_{TB}	I_{RX}	
1	0	88.8	501
2	0	94.1	472

Table 4: Long double rail track circuit, intact infrastructure results

8.2 Infrastructure Failure

Equation used for double rail, Infrastructure Failure analysis:

Maximum permissible interference current, $I_{\text{int\_max}} = \dfrac{5000\sqrt{26.7^2 - I_{TB}{}^2}}{2I_{RX}}$

Number of Tracks	Break Position	Train Position (See Figure 10)	Receiver Current (mA) I_{TB}	Receiver Current (mA) I_{RX}	$I_{\text{int\_max}}$ (mA)
1	1	C	0	377	177 min
1	2	C	0	377	177 min
1	3	C	0	345	193
1	4	C	0	345	193
2	1	C	6.6	274	236
2	2	C	6.6	274	236
2	3	C	5.7	315	207
2	4	C	5.7	315	207

Table 5: Short double rail track circuit, infrastructure failure results

Number of Tracks	Case	Break Position	Train Position (See Figure 10)	Receiver Current (mA) I_{TB}	Receiver Current (mA) I_{RX}	$I_{\text{int\_max}}$ (mA)
1	1	1	C	0	377	177 min
1	2	2	C	0	377	177 min

1	3	3	C	0	345	193
1	4	4	C	0	345	193
2	5	1	C	8.1	274	232
2	6	2	C	8.1	274	232
2	7	3	C	7.0	315	204
2	8	4	C	6.98	315	204
2	9	5	A	0	89.0	750
2	10	6	A	0	94.8	704
2	11	7	A	0	90.5	737
2	12	8	A	0	90.0	741
2	13	9	B	0.5	149	447
2	14	10	B	0.3	226	295

Table 6: Long double rail track circuit, infrastructure failure results

The location of the intermediate bond on a long track circuit has been varied to analyse the relative sensitivity of the track circuit. This has been applied to the same fault conditions as those listed in Table 7 for impedance bond in the middle of the track circuit.

Case	-200m to mid point	-100m to mid point	middle point	+100m to mid point	+200m to mid point
	I_{int_max} (mA)	I_{int_max} (mA)	I_{int_max} (mA)	I_{int_max} (mA)	I_{int_max} (mA)
1	177	177	177 min	177	177
2	177	177	177 min	177	177
3	193	193	193	193	193

4	193	193	193	193	193
5	204	216	232	258	300
6	204	216	232	258	300
7	185	192	204	224	262
8	185	192	204	224	262
9	1068	854	750	693	661
10	688	695	704	716	731
11	703	719	737	758	778
12	786	764	741	722	706
13	407	422	447	484	547
14	275	283	295	316	353

Table 7: Long double rail track circuit, infrastructure failure, sensitivity analysis in respect of the position of the intermediate impedance bond

The cases investigated do not produce the worst case scenario which is confirmed to be when the train shunts the receiver and the broken rail happens to be at the same location.

In case of intact infrastructure, the simulation results for a long double rail track circuit without an intermediate impedance bond have shown only slight increase of the receiver current produced when the train shunt is applied at the transmitter point. Consequently, taking into account the high impedance of the intermediate bond to FS2600 frequencies it has been concluded that sensitivity analyses in respect of different position of the intermediate bond will have a negligible effect on the allowable interference current limits.

8.3 Axle-to-Axle Voltage

Assumed drop-away current, I_{da} = 26.7 mA

Axle-to-axle voltage used in analysis, V_{aa} = 1000 mV

Assumed value of transmitter breakthrough current. I_{TB} = 8 mA (See Table 6)

Equation used for axle-to-axle voltage analysis (see Section 7.4):

Maximum permissible axle-to-axle voltage, $V_{aa\max} = \dfrac{1000\sqrt{26.7^2 - 8^2}}{2I_a}$

Break Position (See Figure ***11***)	Receiver Current I_{aa} (mA)	$V_{aa\_\max}$ (mV)
1	34.5	369
2	34.5	369
3	33.8	377
4	33.8	377

Table 8: Double rail track circuits, axle-to-axle voltage results

9.0 METHODOLOGY FOR DEMONSTRATION OF COMPLIANCE

The longest rack circuit on the route of interest must be determined normally by recourse to the RAR (Railtrack Asset Register). It is expected that this will be shorter than the 1200m track circuit analysed in section 6.4.3.

Since the analysis is based on the most sensitive receiver setting there is no requirement to place any restrictions on future adjustment of FS2600 track circuit receivers on the route.

The acceptability of risks assessed when demonstrating compatibility shall be based on showing that the ALARP (As Low As Reasonably Practicable) principles have been applied. This requires a separate approach depending on whether the hazard occurs in normal operation or in fault condition. The events should be quantified in terms of the rate at which they occur and the overall rate of potential WSF is determined by combining them with track failure data. The exceedance rate for train failures is predicted by the failure analysis.

The rail-break figures and times at risk should be the latest available data and/or statements from Railtrack.

9.1 Normal train - normal infrastructure

Once the longest track circuit on the route is determined the procedure shown in section 7.2 can be followed to evaluate the worst case susceptibility for the track circuits on the route for the case of normal infrastructure. This is then reduced by the safety margin of 3 to determine the train current limits.

For the 1200m track circuit examined in section 8.1 of this document the maximum permissible train interference current is 472 mA for 944 milliseconds. This limit can then be compared to the train emission data. Under all normal operating conditions this limit must not be exceeded.

9.2 Faulty train - normal infrastructure

From the failure analysis carried out the rate at which the train exceed the limit of 472 mA calculated for normal infrastructure in section 8.1 should be determined. The frequency of occurrence, duration and magnitude of the exceedance should be taken into account when constructing mitigating arguments.

9.3 Normal train - faulty infrastructure

The faulty infrastructure scenario as described in 8.2 is used to determine the maximum permissible current value of 177 mA for the given fault scenario. As it can be seen from Table 5 and Table 6 this figure is not dependent on the length of the track circuit. The value is defined for a whole train, independent of the number of units making up the train and constitutes a half of the total allowable interference at the receiver. This figure can be allocated on a per converter basis using root-sum-square addition (the RSS method).

Transient events that may exceed the permissible current curve. These events should be quantified in terms of the rate at which they occur and the overall rate of potential WSFs is determined by combining them with track failure data.

This can be done in the following manner:

For the case of a broken rail occurring on a particular track circuit. The total number of rail breaks per year has to be obtained [18] and the time at risk taken as the average time for a rail break discovered. The number of rail breaks per year for the section of railway of interest must be confirmed by recourse to latest available figures or statements available from Railtrack. The probability of a track circuit having a broken traction rail at any one time is:

Probability of track circuit having broken rail (Ptcbr) =

No of breaks/track miles * TC length(miles)*(time at risk/no of hours per year)

The predicted rate for a train which is standing on a track circuit exceeding the susceptibility level and the traction rail being broken is therefore:

Ptcbr * Exceedance rate (per hour)

This will then give the failure rate in exceedances per hour for normal train and failed infrastructure.

9.4 Faulty train - faulty infrastructure

In this case the train limits and the probability of a track circuit having a broken rail (Ptcbr) are determined in the same way as in section 9.3. The exceedance rate for train failures is predicted by the failure analysis. This failure rate is then used in conjunction with the probability of a track circuit having a broken rail to determine the overall exceedance rate for this case.

9.5 Safety targets

The overall exceedance rate shall be found by adding the rates calculated in sections 9.2, 9.3 and 9.4. This total exceedance rate shall be shown to be ALARP in the context of the safety requirements of the project.

9.6 Axle to axle voltages

The axle to axle voltages should be compliant with the limit defined in section 8.3.

The lowest maximum permissible axle-to-axle voltage is 369 mV, this occurs for track circuits with a break in position 1 or 2 (Figure 11).

APPENDIX A: Data Used for Modelling

PARAMETER	VALUES USED	REFERENCE
Operating Frequency	508.8 Hz (channel 10)	The FS2600 track circuit uses a FSK signal in each of 10 specified operating carrier frequencies. A previous analysis [13] showed that the permissible interference current level was lowest for Channel 10 operation.
Ballast Resistance	Infinite	Worst-case
Rail Parameters		
Internal self inductance of all rails	0.2 μH/m	See Reference 16.
Resistance of all rails	0.25 mΩ/m	See reference 16.
External self inductance of conductor rail	0.48 μH/m	Incorporated in the track circuit models in APPENDIX B: Track Circuit Schematic as from [8].
External self inductance of running rails	0.46 μH/m	Incorporated in the track circuit models in APPENDIX B: Track Circuit Schematic as from [8].

DOUBLE RAIL TRACK CIRCUITS

Track Circuit Length	600 m for low double rail output transmitter (short	Maximum allowable length. See Reference 14, TCH M40, page 1.

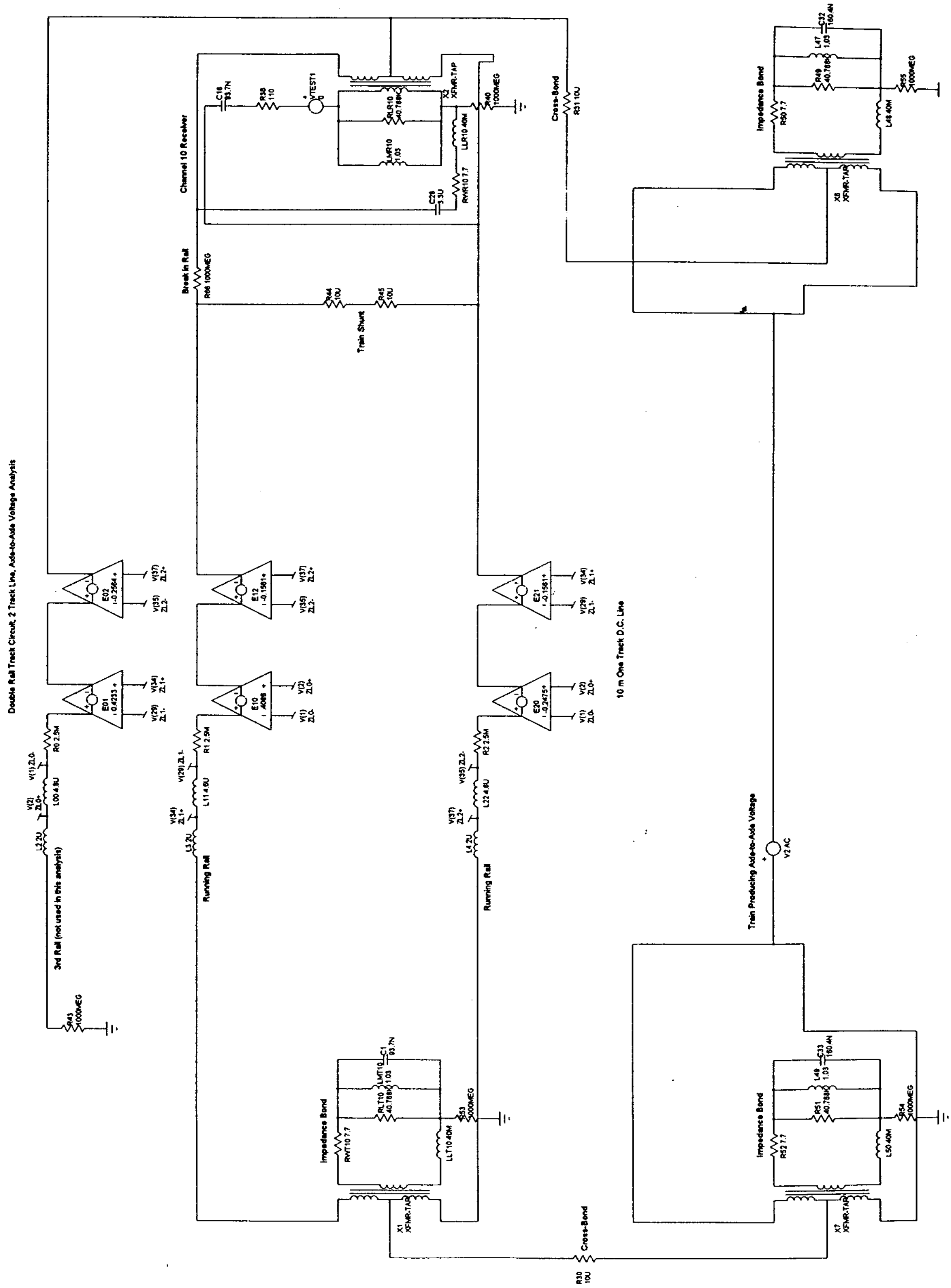
	double rail) 1200 m, with an intermediate bond, for high double rail output transmitter (long double rail)		
Cross-bond Spacing	failure mode analysis: 600 m axle-to-axle voltage analysis: 10m	Spacing of impedance bonds in model. Minimum allowable (worst-case condition). See Reference 14, TCH M40, page 1.	
Transmitter	**Low Double Rail Output**	**High Double Rail Output**	
Output Voltage	120 V (rms)	240 V (rms)	Maximum. See Reference 14, TCH M70, page 4.
Output Resistance	52 Ω	183 Ω	Maximum. See 11, section 4.2.2.2.
Capacitor Value	None	None	See 11, section 2.2.
Receiver			
Gain	See explanation in Appendix 2.		
Input Resistance	110 Ω		Average value. See Reference 11, Table 6.

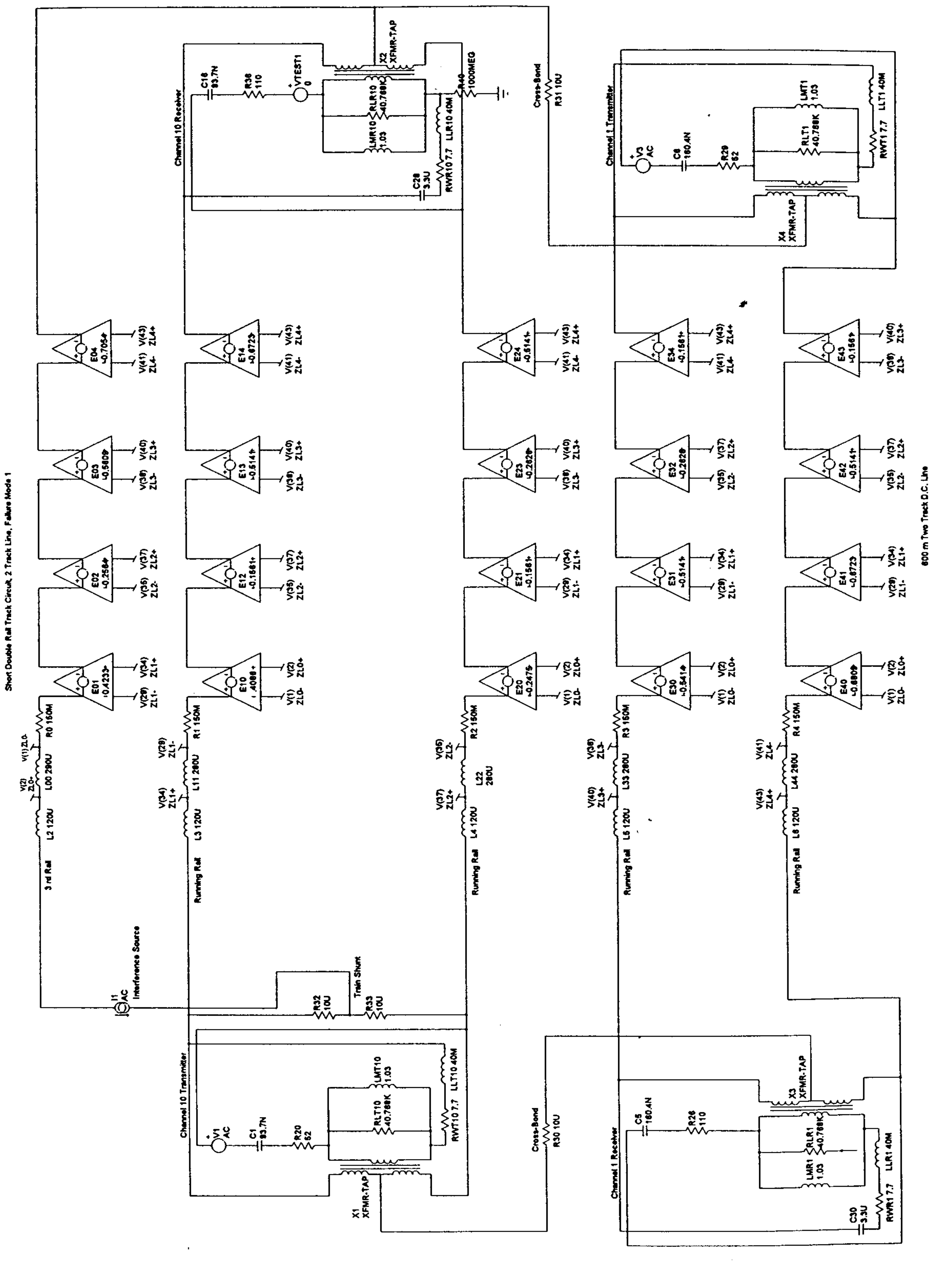
Capacitance	3.3 μF	Average value. See Reference 11, Table 6. Optional 12.3 μF capacitor not included. (See transmitter capacitance.)
Rail to rail drop-away voltage	1.6 V (rms)	See 11, section 4.1.2.1.

Impedance Bond – type S

Tuning Capacitor	Channel 10: 93.7 nF Channel 1: 160.4 nF	Corresponding value to the Westinghouse Signals part number in Reference 7.
Turns Ratio of impedance bond	1:56 (1+1:112)	See Reference 11, Figure 2.
Loss Resistance	40.768 kΩ	See Reference 14, page 3.
Other Impedance Bond Parameters	MagnetisingInductance: 1.03 H Winding Resistance: 7.7 Ω Leakage Reactance: 40 mH	Calculated from measurements on and actual S2 type impedance bond. See Reference 7.

APPENDIX B: Track Circuit Schematics




Short Double Rail Track Circuit, 2 Track Line, Failure Mode 1
600 m Two Track D.C. Line
Channel 10 Receiver
Channel 1 Transmitter
Channel 10 Transmitter
Channel 1 Receiver
Interference Source
Train Shunt
Cross-Bond
3 rd Rail
Running Rail

APPENDIX C: ASSUMPTIONS

Assumptions: configuration, supporting references and infrastructure failure modes

General

1 Only double rail configuration of FS2600 track circuit has been considered, as it is the only one used on RT controlled infrastructure.

2 The running rails are assumed to be type BS113A, normal grade. Check rails are excluded. The sleepers are conventional concrete or wooden type. The track is assumed to be of standard gauge.

3 Effects due to the length of the train occupying the track circuit are ignored (i.e. a point source is assumed).

4 The train producing axle-to-axle voltage produces the voltage at a spacing equal to the cross-bond spacing. Current flow in the train and track under the train are not considered.

5 The analysis was carried out for a channel 10 track circuit (i.e. 508.8 Hz), with a channel 1 (388.8 Hz) track circuit on the adjacent track.

6 The interference source was always positioned to achieve the maximum effect on the track circuit receiver.

7 Only one train was ever assumed to be on the track, drawing current, and therefore no current would flow in the other third rail in case of double track. Currents produced by other trains in the area are allowed for by the interference budget.

Configuration

8 Impedance of the cables used for cross-bonding was represented by 10 $\mu\Omega$ resistances.

9 An internal self inductance of 0.2 μH/m was used for all rails.

10 A resistance of 0.25 Ω/m was used for all rails.

11 An external self inductance of 0.48 μH/m was used for the conductor rail.

12 An external self inductance of 0.46 μH/m was used for the running rails.

Infrastructure failure modes

13 The transmitter is assumed to be active when the interference is generated, to correctly simulate the real situation.

14 The voltages produced by transmitter breakthrough and interference are combined using root mean square addition.

15 The train interference allowance used for Intact Infrastructure Case is a third.

16 The train interference allowance used for Faulty Infrastructure Case is a half.

17 The train interference allowance used for wrong side failure caused by axle-to-axle voltage is a half for double rail track circuits.

18 It is assumed that the interference generated current at the receiver is proportional to the interference current, for intact infrastructure.

APPENDIX D: HAZARDS

Interference current in traction return current

This hazard is covered by the assessment as presented.

Axle to Axle or Longitudinal voltage

This hazard is covered by the assessment as presented.

Direct induction into track mounted or trackside components

Alternating magnetic fields at FS2600 frequencies need not be considered for possible induction effects since very high interference field levels would be required to have a significant effect.

Particular transient events

No single type or source of transient has been identified as posing a particular risk to FS2600 track circuits.-

Wheelset impedance

The maximum wheelset impedance is not quoted in this documen

Vehicle geometry (overhang, wheelbase etc..)

This hazard is not covered by this document.

APPENDIX E: MITIGATING FACTORS

Pessimistic assumptions

Train occupying the track circuit under consideration was modelled as a point source.

A worst case train position is considered in the analysis.

The effects of ballast resistance and inter-rail capacitance were ignored.

Interference budget accounts for arithmetic addition of the interference currents from all sources. In practice it is unlikely that the interference sources would be producing the same signal, in phase.

Transmitter breakthrough is taken into account into the maximum current allowable at the receiver when there is a train on the track circuit. As there is no synchronising mechanism between transmitter and train, it is incredible that the interference produced by the train could be switching between the two required frequencies at exactly the same rate as the track circuit. Therefore it is assumed that these can be combined using root-mean-square addition.

Impedance bonds' saturation effect is not considered.

The permissible interference current level is lower for operation at Channel-10 carrier frequency of 508.8 Hz [13].

The distance between the axles of the train producing an axle-to-axle voltage, is assumed to be the same as the distance between the cross bonds (10m).

Background/coincident sources of interference covered by train interference allowance

Other rolling stock, including trains producing axle-to-axle voltage.

Background current from other sources.

APPENDIX F: SOURCES OF TECHNICAL WORK

The technical work contained in this document is taken from a number of sources:

- The Class 92 EMC safety case, produced by AEA Technology Rail, ISAs were WS Atkins Rail (secretariat reviewer) and Mott MacDonald.
- The Class 365/5 Networker Express EMC safety case, produced by AEA Technology Rail, ISA was Mott MacDonald.

The Juniper EMC safety case, produced by AEA Technology Rail, ISA was WS Atkins Rail.

RAILTRACK

BRIEFING NOTE

RT/E/C/50001 to RT/E/C/50019 **Issue 1**

Methodology for the demonstration of electrical compatibility between rolling stock and infrastructure

Issue Date: February 2003

Background / Synopsis

This suite of Codes of Practice is intended to assist train-builders, TOCs, and ROSCOs in the production of Route Acceptance Safety Cases for electric trains, by describing a methodology for demonstrating compatibility with EE&CS systems on Railtrack's electrified routes.

The electrical susceptibility of the key items of infrastructure is described and characterised, and recommended methods for demonstrating compatibility between rolling stock and infrastructure are given.

Key Changes / Compliance

This is a new suite of documents. As the documents are Codes of Practice, compliance is not mandatory.

Implementation

The information contained in the documents may be used with immediate effect. The various documents in the suite will be published progressively during 2003.

Of Interest To:

Authors of Route Acceptance Safety Cases.

Name and Title of Author

David Wilkinson
Technical & Safety Advisor